

A270 – Z_3 Sector Structure and Hadron Mass Correction in T^4/Z_3

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Contents

.1	Purpose and Status	1
.2	Starting Point: Bulk Exponent 36	1
	The Key Relation	1
	Statistical Safeguard	1
.3	T^4/Z_3 Orbifold Structure	2
	Fixed-Point Geometry	2
	Quark Localization and Confinement	2
	Sector Exponents	2
.4	Empirical Anchor: the Pion	3
.5	Consistency Evidence	3
	Koide Relation	3
	K_{frak} Distribution in the Corpus	3
.6	Prediction: Baryon Step	3
.7	Open Points and Falsification	4
.8	Notation	4
.9	References	5

.1 Purpose and Status

[STIPULATION] This document investigates the sector structure of the T^4/Z_3 orbifold with regard to a structural distinction between elementary particles (leptons) and composite particles (hadrons). Central result: the three Z_3 sectors assign to each particle class a topologically fixed K_{frak} exponent — leptons 0, mesons +1, baryons +2 — in addition to a common bulk exponent 36.

Status: hypothesis at consistency level [K]. The meson anchor is passed via the pion binding (A155). The baryon step is a prediction; its direct test requires a geometric baryon base mass (open).

.2 Starting Point: Bulk Exponent 36

The Key Relation

[K] Two independently determined FFGFT quantities meet in one number. First, the D4 packing density $\pi^2/16$, whose reciprocal is

$$\frac{16}{\pi^2} \approx 1.62114.$$

Second, the scale-recursion depth from Doc. 275,

$$k^* = \frac{\ln \varphi}{\xi} \approx 3609.1 \quad \Rightarrow \quad \frac{k^*}{100} = 36.09.$$

Numerically:

$$K_{\text{frak}}^{-36} = \left(\frac{75}{74}\right)^{36} \approx 1.62130 \approx \frac{16}{\pi^2} \quad (\Delta = 0.010 \%).$$

Statistical Safeguard

[K] Monte-Carlo check (100 000 trials): the hit probability $\leq 0.010 \%$ for a random target in the value range is 1.56% per candidate. Of 21 tested physical constants (π , e , φ , $\sqrt{2}$, $\sqrt{3}$, $\ln 2$, ...) *only* $16/\pi^2$ hits at $\leq 0.010 \%$; the next-best candidate lies at 0.066% . The independent support by $k^*/100$ (Doc. 275 was determined without reference to masses) considerably weakens the chance hypothesis. P35 remains noted: a forward derivation of the identity $K_{\text{frak}}^{-36} = 16/\pi^2$ does not exist.

[STIPULATION] The exponent 36 is henceforth called the *bulk exponent*: the common fractal correction of the untwisted sector in which all particles live.

.3 T^4/Z_3 Orbifold Structure

Fixed-Point Geometry

[B] Z_3 acts on $T^3 = S_x^1 \times S_y^1 \times S_z^1$ by cyclic permutation:

$$(x, y, z) \xrightarrow{g} (y, z, x) \xrightarrow{g} (z, x, y) \xrightarrow{g} (x, y, z)$$

Fixed points: $x = y = z = k \cdot L/3$, $k = 0, 1, 2$ — exactly **3 fixed points**, isomorphic to Z_3 .

Local geometry at each fixed point:

Direction	Vector	Z_3 sector
$\mathbf{e}_0 = (1, 1, 1)/\sqrt{3}$	invariant	sector 0 (untwisted)
$\mathbf{e}_1 = (1, -1, 0)/\sqrt{2}$	phase ω	sector g (twisted)
$\mathbf{e}_2 = (1, 1, -2)/\sqrt{6}$	phase ω^2	sector g^2 (twisted)

At each fixed point: 1 untwisted + 2 twisted degrees of freedom — exactly the Z_3 sector structure $\{e, g, g^2\}$.

Quark Localization and Confinement

[H] In FFGFT, $\tilde{T} \cdot m = 1$ holds; the mass m sits on S_m^1 , one of the three spatial circles. Under Z_3 permutation, S_m^1 has two images on the other circles.

Elementary particle (lepton): sits on S_m^1 alone, transforms as a 1-dimensional Z_3 representation, remains in the untwisted sector 0.

Composite particle of 3 constituents (baryon): constituent 1 on S_x^1 , constituent 2 on S_y^1 , constituent 3 on S_z^1 . The localization condition "all 3 at the same spatial point" reads:

$$x = y = z \quad \Leftrightarrow \quad \text{fixed-point condition}$$

Quark confinement in T^4/Z_3 is thereby *topologically enforced*: free quarks would violate the fixed-point condition. Leptons (1 circle) have no fixed-point condition and are free.

Sector Exponents

[H] K_{frak} is a property of the sector partition function. Each of the 3 Z_3 sectors contributes K_{frak}^1 . The bulk exponent 36 covers the untwisted sector (already contained in the ladder formulas). Twisted sectors contribute an additional K_{frak}^1 each. Hence the sector table:

Particle class	Const.	Active sectors	Exponent
Lepton (e, μ, τ)	1	$\{e\}$	$36 + 0$
Meson ($q\bar{q}$)	2	$\{e, g\}$	$36 + 1 = 37$
Baryon (qqq)	3	$\{e, g, g^2\}$	$36 + 2 = 38$

The exponents are topologically fixed: $Z_3 = \{e, g, g^2\}$ has exactly two non-trivial elements, independently of any mass.

.4 Empirical Anchor: the Pion

[K] A155 introduces the additive formula $m_M = m_{q_1} + m_{q_2} + \Lambda_{\text{QCD}} \cdot K_{\text{frak}}^{\text{neff}}$ for mesons. Evaluating the pion binding yields (A155):

$$\boxed{n_{\text{eff}}(\pi^0) = 37.79} \quad n_{\text{eff}}(\pi^\pm) = 36.62.$$

Both values lie within the window of the sector prediction $36 + 1 = 37$ for mesons. **The meson step of the sector table is thereby empirically anchored at the pion** — the exponent was determined in A155 (from the legacy-corpus formula) independently of this sector argument.

The domain of validity of the additive formula (only binding $< \Lambda_{\text{QCD}}$; kaon, ρ , ω structurally outside) and the GMOR extension are treated in A155.

.5 Consistency Evidence

Koide Relation

[K] The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}$$

holds exactly for leptons (A110: the geometric ladder predicts Q within 0.16 % of $2/3$ without using Koide; the experimental value hits $2/3$ exactly with $|\Delta| \approx 2 \times 10^{-6}$).

For quarks, according to A150, *no* corresponding structure exists:

“Second, there are six quarks instead of three leptons, but no ordering corresponding to the Z_3 structure that would map them onto three eigenvalues of one operator. The circulant plane of A110 has no counterpart here.” (A150, §Quarks)

This is consistent with the sector hypothesis — and sharpens it: the Z_3 circulant (A110), which carries the Koide structure of the leptons, is the algebra of the untwisted sector. Leptons in the untwisted sector $\rightarrow$ circulant acts $\rightarrow$ Koide holds. Quarks couple to twisted sectors $\rightarrow$ the circulant ordering has “no counterpart” (A150) $\rightarrow$ no Koide analogue exists. The sector hypothesis thus supplies a candidate for the *reason* of the structural deficit recorded in A150.

K_{frak} Distribution in the Corpus

[K] K_{frak} appears in the corpus base formulas exactly where the sector table demands it:

Class	K_{frak} in base formula?	Sector table
Leptons	no ($m_L = r \xi^p v$, A100)	untwisted: no extra
Mesons	yes ($\Lambda \cdot K_{\text{frak}}^n$, A155)	+1 sector
Baryons	yes; corrected legacy calculation with anchor $\pi^2/2$ (Doc. 190, R62; $\Delta = 0.05\%$)	+2 sectors

.6 Prediction: Baryon Step

[H] From the sector table, the exponent for baryons is $36 + 2 = 38$:

$$K_{\text{frak}}^{-38} = 1.6654.$$

The corresponding statement: a geometric baryon base mass (before sector correction) relates to the observed one by K_{frak}^{-38} ; equivalently, the baryon exponent lies exactly one unit above the meson value measured at the pion.

Direct test open: it requires an independently derived geometric baryon base mass, which does not exist in the corpus (A150: the quark sector is descriptive classification, no predictive quantity). Until then, the baryon step rests on the topology (two twisted sectors) and the pion anchoring of the meson step.

.7 Open Points and Falsification

1. **Forward derivation of $K_{\text{frak}}^{-36} = 16/\pi^2$** (P35): the identity is supported numerically and via $k^*/100$, but not derived.
2. **Baryon sector:** the legacy calculation is corrected by register entry (Doc. 190, R62). A fully geometric candidate for the proton mass — $m_p = (\pi^3/12) N_c (1 + \alpha_s) e^{-3\xi/4} \Lambda_{\text{QCD}}/2$, every factor interpreted, $\Delta = +0.21\%$ absorbable — is kept in A155 §Baryon Candidate ([S]; forward derivation open). The direct test of the +2 step awaits this derivation.
3. **$\pi^\pm/\pi^0$ difference** (36.62 vs. 37.79): does the EM self-energy carry a sector contribution of its own? (A155)

Falsification hierarchy:

1. Meson exponent $36 + 1$: confirmed at the pion (37.79, A155); extension to the kaon via the GMOR form in A155 as [S] candidate.
2. Koide analogue for quarks: must *not* hold — consistent with A150 ("circulant plane of A110 has no counterpart") ✓
3. Baryon exponent $36 + 2$: falsifiable as soon as a geometric baryon base mass exists.

.8 Notation

Symbols used throughout the A-series are explained in A010. Specific to this document:

Symbol	Meaning
$Z_3 = \{e, g, g^2\}$	cyclic group; g : permutation $(x, y, z) \rightarrow (y, z, x)$
$\omega = e^{2\pi i/3}$	primitive third root of unity (twist phase)
sectors $0, g, g^2$	untwisted sector; the two twisted sectors (eigenvalues $1, \omega, \omega^2$ of the permutation)
$\mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2$	eigenbasis at the fixed point: invariant direction $(1, 1, 1)/\sqrt{3}$; twisted directions
fixed point	point with $x = y = z$ on T^3 ; three of them ($k \cdot L/3$)
bulk exponent 36	common K_{frak} exponent of the untwisted sector; $K_{\text{frak}}^{-36} \approx 16/\pi^2$
sector exponent	addition per particle class: lepton +0, meson +1, baryon +2
$k^* = \ln \varphi / \xi$	scale-recursion depth to $1/\varphi$ (Doc. 275); $k^*/100 = 36.09$
$\pi^2/16$	D4 packing density; reciprocal $16/\pi^2$
n_{eff}	empirical binding exponent from A155 (pion anchor)
Q	Koide quantity $(m_e + m_\mu + m_\tau) / (\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2$

Verification script: `python/A_Serie_Skripte/a270_z3_sektoren.py` — all numerical statements of this document are checked there with assertions.

.9 References

Document	Reference
A015	origin of ξ , Casimir confirmation
A100	lepton ladder (r_i, p_i)
A110	circulant and Koide: $Q = 2/3$, Z_3 algebra of the leptons
A150	quark ladder, lepton/quark distinction
A155	meson formula: domain, pion anchor, GMOR; baryon candidate
Doc. 190	P35, P40; register R62 (baryon correction, legacy corpus)
Doc. 275	$k^* \approx 3609$, scale recursion, $1/\varphi$ passage
Doc. 230/231/232	T^4/Z_3 Hilbert-space bridge